

Where Severe Crashes Concentrate: A Spatial Analysis of Killed-and-Severe-Injury Collisions in Redondo Beach and Torrance, California

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Introduction

Traffic crashes are among the most spatially concentrated public-safety problems a city faces: a small share of streets typically accounts for a disproportionate share of serious injuries and deaths. This concentration is the premise behind Vision Zero, the road-safety framework adopted by Los Angeles County and many of its cities, which directs engineering investment toward a high-injury network rather than spreading resources evenly across every road. Accordingly, this report focuses on killed-and-severe-injury (KSI) collisions rather than all reported crashes (Ferrier, 2018). Mapping every reported collision largely tracks where traffic volume is highest; isolating KSI crashes concentrates attention on the outcomes traffic safety policy exists to address.

Redondo Beach and Torrance were selected for both practical and analytical reasons. These adjacent South Bay municipalities share a boundary while exhibiting different street patterns, Redondo Beach's coastal grid against Torrance's more regular inland arterial network, and together they provide a connected street network large enough to compare high- and low-injury corridors. Redondo Beach also supports dense pedestrian activity in areas such as Riviera Village, where heavy foot traffic coincides with a street environment that can make safe crossing difficult.

This study asks where KSI crashes concentrate across the two cities and whether the highest-injury corridors coincide with land uses that generate pedestrian activity. Kernel density estimation identifies areas of elevated concentration, a spatial join attributes crashes to named corridors, and quarter-mile buffers around schools, parks, and transit stops establish proximity to pedestrian generators. The intent is to distinguish streets that are merely busy from streets that are genuinely dangerous to people on foot, which is the framing a municipal public works department would need in order to act.

Methods

Study area. The study area covers Redondo Beach and Torrance in the South Bay region of Los Angeles County (Figure 1). Torrance was included to enlarge the sample of serious crashes and to allow comparison of high- and low-injury corridors across a connected street network.

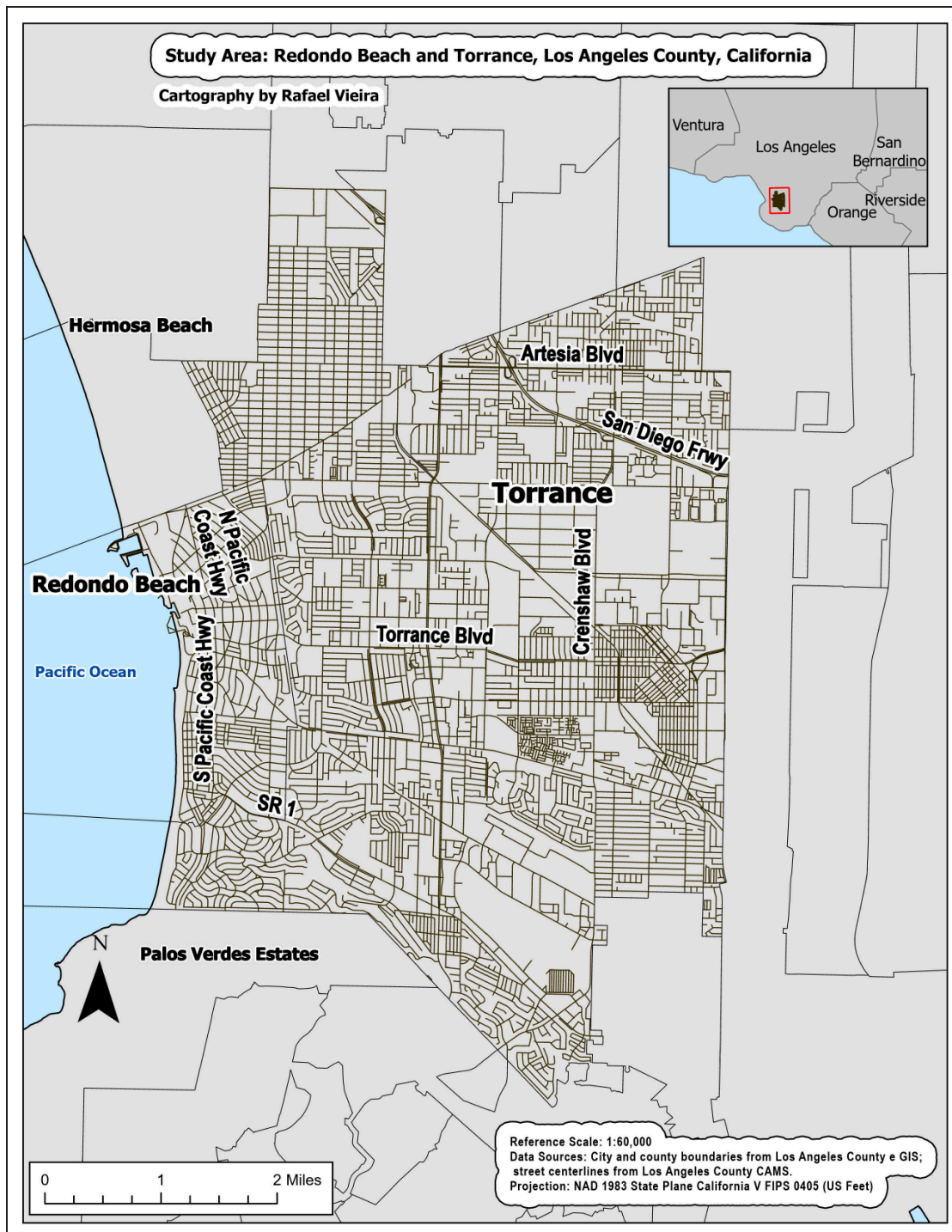


Figure 1. Study area: the cities of Redondo Beach and Torrance, Los Angeles County, California.

Data sources. This analysis draws on four datasets, none used in prior course assignments. Crash records were obtained from the California Statewide Integrated Traffic Records System (SWITRS), accessed through UC Berkeley’s Transportation Injury Mapping System, covering 2021 through 2025 (SafeTREC, 2026). Street centerlines came from the Los Angeles County Countywide Address Management System (Los Angeles County, 2026). Transit stop locations were compiled

from the published General Transit Feed Specification (GTFS) feeds of LA Metro, Torrance Transit, and Beach Cities Transit. School and park locations were digitized manually in ArcGIS Pro, referencing a base map. All datasets were accessed in July 2026.

Data preparation. Each dataset required preparation before analysis. The crash records were delivered as flat CSV files without spatial geometry, so each was converted to point features using XY Table to Point with the longitude and latitude fields and a GCS_WGS_1984 coordinate system. Because Redondo Beach and Torrance were downloaded as separate queries, the two point layers were combined with the Merge tool. A review against the street network revealed a small number of records geocoded offshore in the Pacific; these were clearly erroneous and were removed. The transit stops underwent the same conversion, and the three operators' layers were likewise merged so that gaps in any one operator's coverage would not distort the analysis. All layers were then clipped to the two-city boundary using Pairwise Clip. A severity filter on the collision severity field returned 99 KSI crashes.

Spatial analysis. Three spatial processes were applied to the prepared data. First, a kernel density surface was generated from the 99 KSI crash points. Second, a spatial join assigned each crash to its nearest street centerline, allowing high-injury concentrations to be named as specific corridors rather than diffuse clusters; summary statistics then produced a ranked count per corridor. Third, quarter-mile buffers were generated around schools, parks, and transit stops, a radius transit planners treat as the walkable catchment of a bus stop (WMATA, 2014); these were dissolved into a single proximity zone and overlaid against the crash points using select-by-location.

Results

Of the 99 KSI crashes recorded between 2021 and 2025, 8 were fatal and 91 resulted in severe injury, so fatalities accounted for roughly 8 percent of the most serious outcomes. Both categories represent the outcomes Vision Zero aims to eliminate, and together they define the high-injury pattern examined below (Table 1, Figure 2).

Table 1. KSI crashes by severity, Redondo Beach and Torrance, 2021–2025.

Severity	Count	Percent
Fatal	8	8.1%
Severe injury	91	91.9%
Total	99	100%

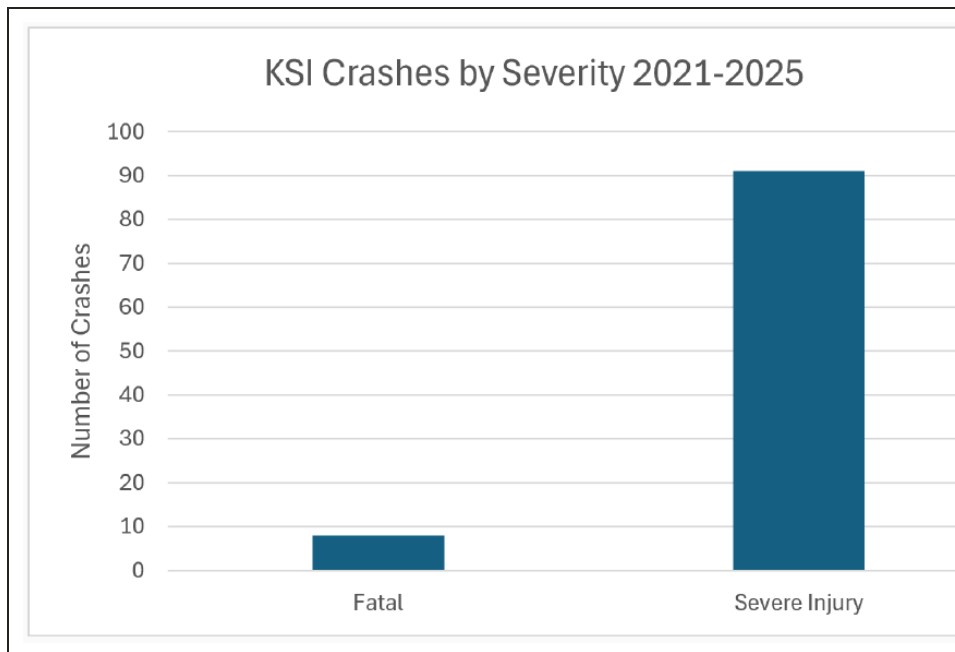


Figure 2. KSI crashes by severity, 2021–2025.

The kernel density surface (Figure 3) shows severe crashes concentrated heavily along the coast, with the densest cluster in western Redondo Beach near Riviera Village and a secondary concentration further north. Inland Torrance shows markedly lower KSI density across most of its street network.

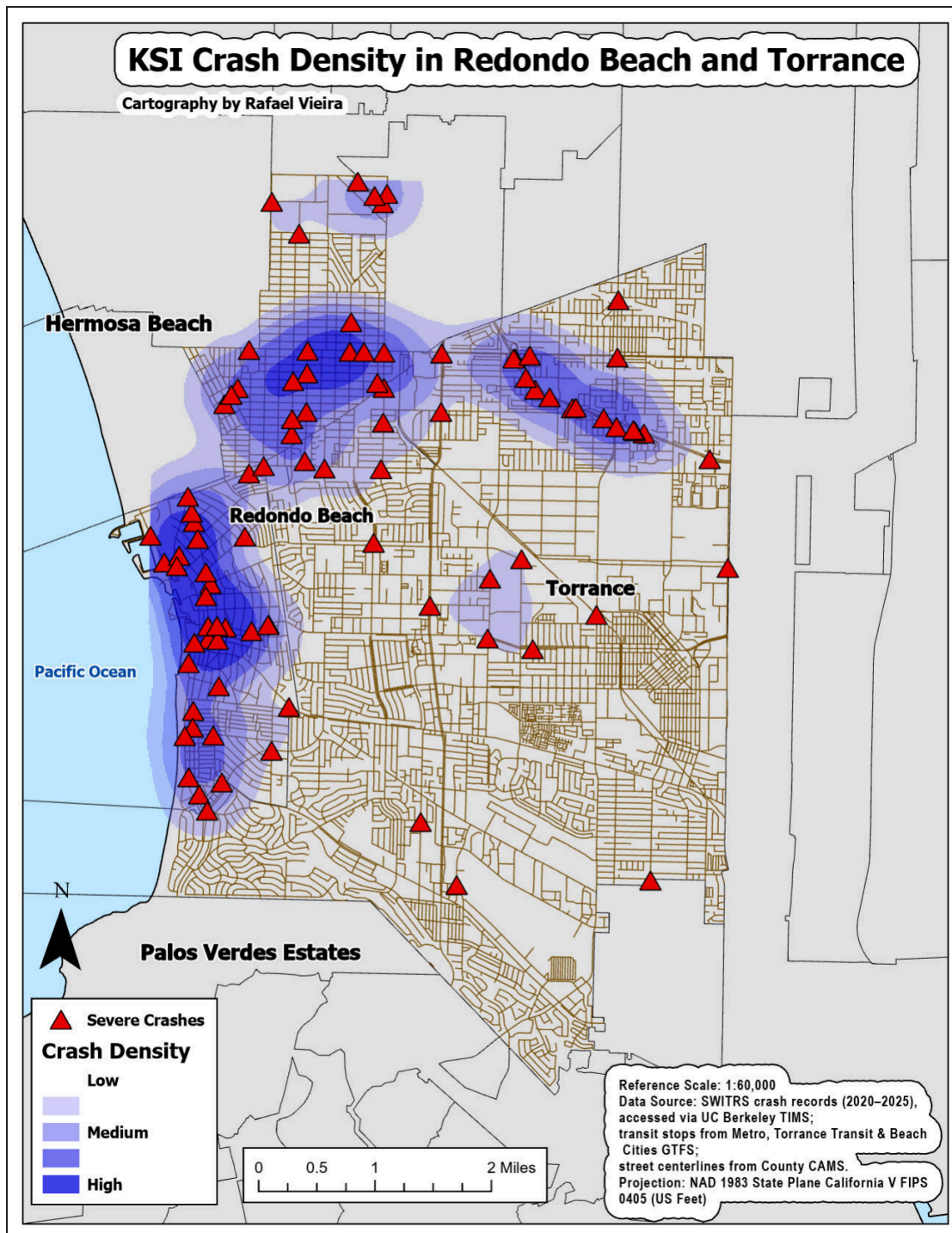


Figure 3. Kernel density of KSI crashes in Redondo Beach and Torrance.

The spatial join identified the corridors carrying the highest KSI counts (Figure 4, Figure 5, Table 2). The San Diego Freeway recorded the single highest count at 12, but functions as a limited-access facility rather than a walkable street, and its crash mechanism differs fundamentally from that of surface streets. Among surface corridors, Artesia Boulevard carried the most severe crashes at 10, followed by Pacific Coast Highway at 8 once its separately listed northern,

southern, and State Route 1 segments are combined. Crenshaw Boulevard and Torrance Boulevard followed at 3 each. In total, 58 corridors recorded at least one KSI crash, but 41 recorded only one, confirming the concentration the Vision Zero framework predicts.

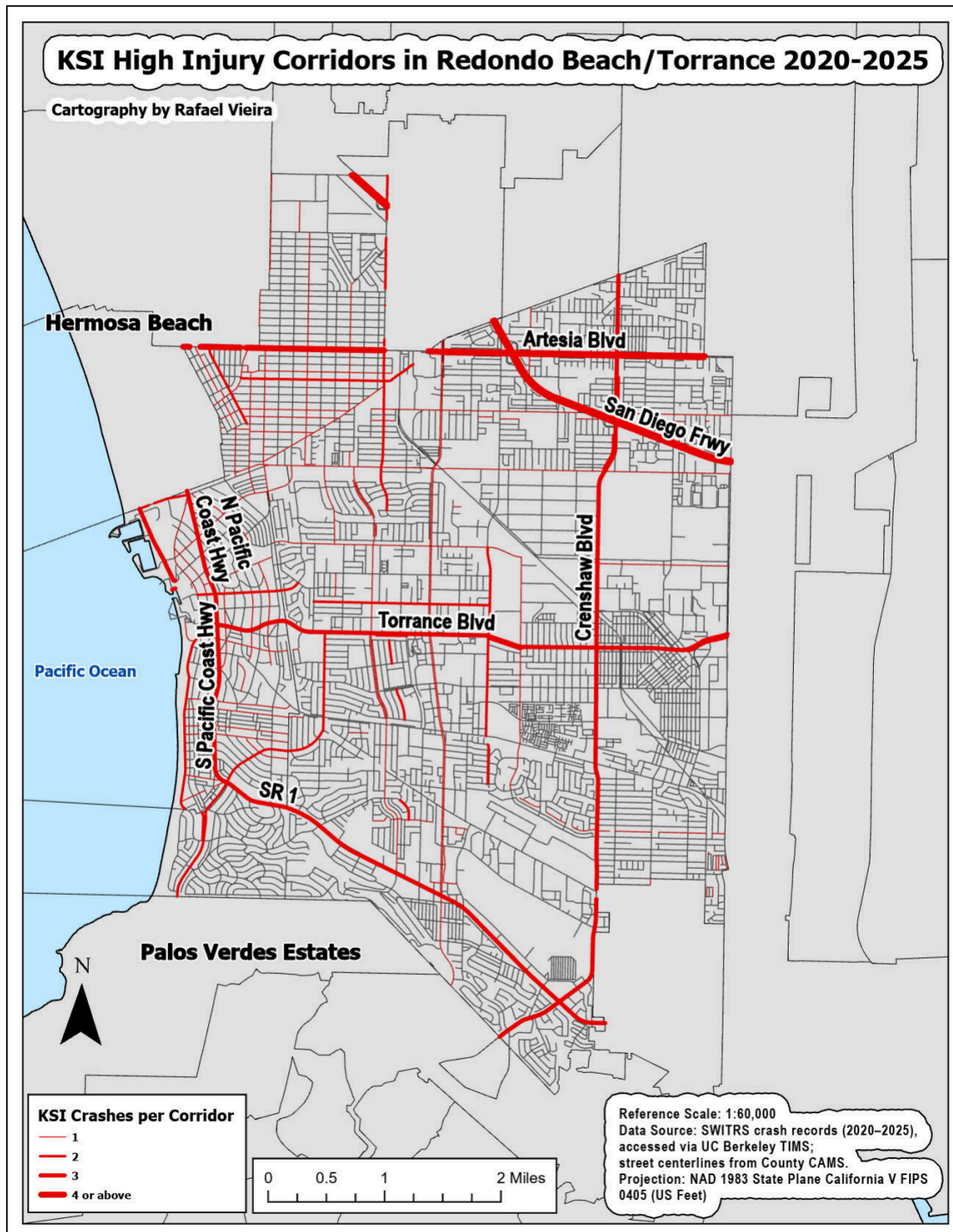


Figure 4. High-injury corridors by KSI crash count, Redondo Beach and Torrance.

Table 2. Top five corridors by KSI crash count, 2021–2025.

Corridor	KSI crashes
San Diego Freeway*	12
Artesia Boulevard	10
Pacific Coast Highway	8
Crenshaw Boulevard	3
Torrance Boulevard	3

* Limited-access freeway; not directly comparable to surface streets.

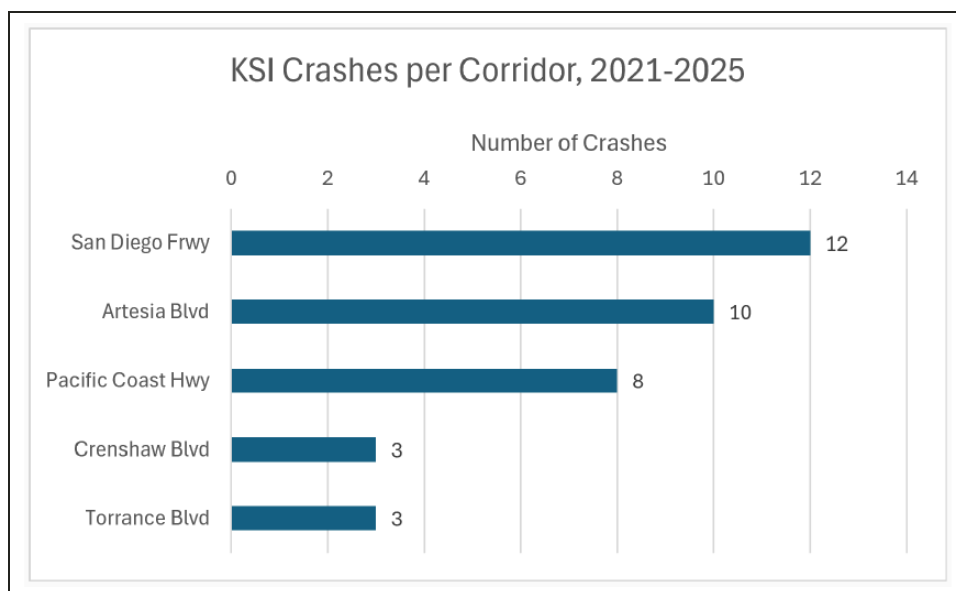


Figure 5. KSI crashes per corridor, 2021–2025. The San Diego Freeway is a limited-access facility and is not directly comparable to surface streets.

The proximity analysis found that 82 of the 99 KSI crashes, or 82.8 percent, occurred within a quarter mile of at least one pedestrian generator, leaving 17 outside that zone (Figure 6, Table 3). By generator type, 77 crashes (77.8 percent) fell within a quarter mile of a transit stop, 22 (22.2 percent) of a school, and 15 (15.2 percent) of a park. These categories are not mutually exclusive, as many crashes fell near more than one generator type (Figure 7).

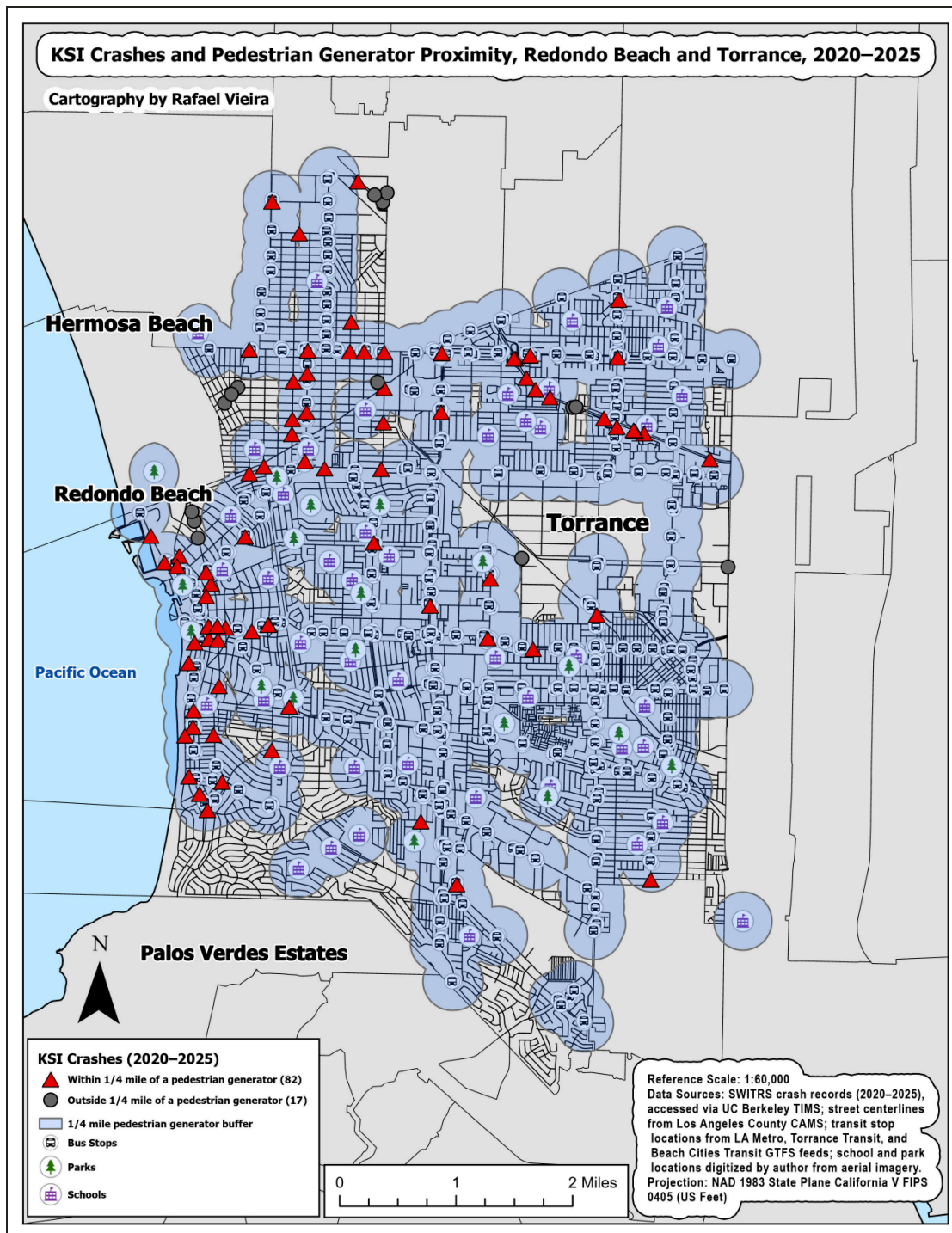


Figure 6. KSI crashes by proximity to pedestrian generators, Redondo Beach and Torrance.

Table 3. KSI crashes within a quarter mile of pedestrian generators, 2021–2025.

Pedestrian generator	KSI crashes within ¼ mile	Percent of total
Transit stops	77	77.8%
Schools	22	22.2%
Parks	15	15.2%
Any generator	82	82.8%

Categories are not mutually exclusive; many crashes fell within ¼ mile of more than one generator type.

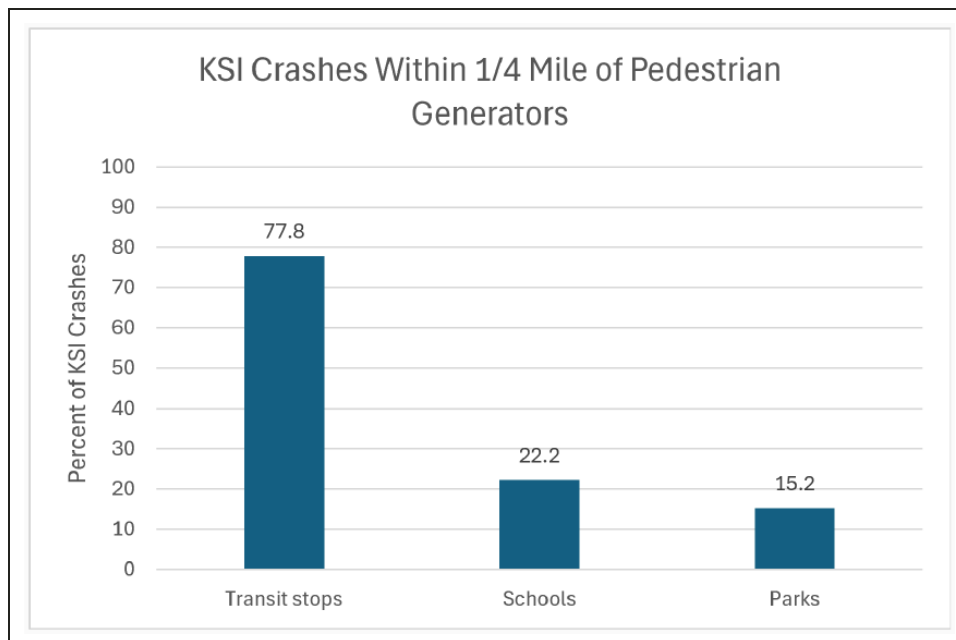


Figure 7. Percentage of KSI crashes within a quarter mile of each pedestrian generator type, 2021–2025. Categories are not mutually exclusive.

Discussion

The clearest analytical result of this study is that its two measures of danger do not identify the same places. The kernel density surface peaks on the coast of Redondo Beach, while corridor counts peak inland on Artesia Boulevard. This is not a contradiction but a reflection of the fact that density and count measure different things: the coastal cluster is dense because many severe crashes are packed into a small, walkable area, while Artesia ranks highly because crashes are spread along miles of continuous arterial.

That difference is likely rooted in road design and exposure. Wide, fast arterials such as Artesia Boulevard generate severe outcomes primarily through vehicle speed, since the same collision produces far worse injuries at higher impact speeds. Dense coastal areas such as Riviera Village generate them primarily through exposure, meaning large numbers of people on foot crossing and moving alongside traffic. A safety strategy built on only one of these measures would systematically miss the other type of location.

The proximity analysis supports the exposure interpretation but requires careful reading. Bus stops do not cause crashes; they mark where people are walking, waiting, and crossing near busy roads. Notably, Artesia Boulevard, the highest-count surface corridor, also carries a high density of bus stops, so corridor rank and transit proximity converge there. School proximity, by contrast, showed a weaker and more evenly distributed association, suggesting that school density alone does not predict where severe crashes occur.

Several limitations qualify these findings. The transit result is inflated by the sheer number of bus stops: because they blanket the street network, almost any crash location falls within a quarter mile of one, which limits that variable's discriminating power. A better-weighted indicator, such as boardings per stop, would sharpen the test. Freeway crashes were retained rather than excluded, a departure from standard high-injury-network methodology, which is why they are treated separately above. KSI counts are also not normalized by traffic or pedestrian volume, and five years of data yields small counts on individual corridors, where a single crash can shift the ranking.

For a planner, these findings distinguish two safety problems that call for different responses. The coastal Redondo Beach cluster reflects high pedestrian exposure and would be addressed through measures such as improved crossings, signal timing, lighting, and reduced speeds. The high counts on arterials such as Artesia Boulevard reflect road design and vehicle speed, calling for interventions such as lane reconfiguration and turn treatments. Further work could test road width, posted speed, traffic volume, and surrounding land use to explain why.

Conclusion

This study asked where killed-and-severe-injury crashes concentrate across Redondo Beach and Torrance and whether those concentrations align with pedestrian-generating land uses. Severe crashes cluster most densely along the coast of Redondo Beach, the highest surface-street corridor counts fall on wide arterials such as Artesia Boulevard, and 82.8 percent occur within a quarter mile of a pedestrian generator. That the densest cluster and the highest-count corridors are not the same place is the study's central insight, and it suggests that safety investment must be targeted to the mechanism at work in each location.

References

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